

Effect of Cogent db, a herbal drug, on serum and tissue lipid metabolism in experimental hyperglycaemic rats

G. Saravanan and L. Pari

Department of Biochemistry, Faculty of Science, Annamalai University, Annamalaiagar, Tamil Nadu-608 002, India

Aims: We have previously reported the antidiabetic effect of Cogent db. The present study with alloxan-induced hyperglycaemic rats is focused on the mechanism of action, specifically on the activity of hepatic lipogenic enzymes, serum and tissue lipids.

Methods: Male Wistar rats body weight of 180–200 g (six normal and 18 diabetic rats) were used in this study. The rats were divided into four groups after the induction of alloxan diabetes: normal rats; diabetic control; diabetic rats given Cogent db (0.45 g/kg body weight); diabetic rats given glibenclamide (600 µg/kg body weight). After 40 days treatment, fasting blood glucose, plasma insulin, activities of hepatic lipogenic enzymes, serum and tissue lipids were determined in normal and experimental animals.

Results: Oral administration of Cogent db for 40 days resulted in significant reduction in blood glucose, serum and tissue (liver and kidney) lipids, whereas the level of plasma insulin and the activity of hepatic lipogenic enzymes were significantly increased in alloxan diabetic rats. Similar studies using glibenclamide have been conducted to compare the mode of action of these two drugs.

Conclusions: Thus our study shows that Cogent db exhibits a strong antihyperlipidaemic effect, which could exert a beneficial action against macrovascular complications (cardiovascular disease) associated with diabetes mellitus.

Keywords: alloxan diabetic rats, plasma insulin, Cogent db, blood glucose, lipid metabolism, malic enzymes, glucose-6-phosphate dehydrogenase

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Introduction

Diabetes mellitus (DM) is a common disorder associated with markedly increased morbidity and mortality rate. DM can be defined as a group of metabolic diseases characterized by chronic hyperglycaemia resulting from defects in insulin secretion, insulin action or both, resulting in impaired function in carbohydrate, lipid and protein metabolism [1].

Macrovascular complications associated with type 2 diabetes mellitus are the major contributors to the increased morbidity and mortality in patients with this

disorder. In developed countries, there is a two- to four-fold increase in the risk of cardiovascular disease in diabetic patients compared with non-diabetic patients [2]. The reasons for this increased risk are only partly understood and probably involve several factors, including hyperglycaemia and the existence of co-morbidity factors, e.g. hyperinsulinaemia, hypertension and obesity [3]. In addition, type 2 diabetes mellitus adversely affects the plasma lipid profile, increasing levels of atherogenic lipids such as low-density lipoprotein (LDL) and very-low-density lipoprotein (VLDL), but

Correspondence:

Dr L. Pari, Department of Biochemistry, Faculty of Science, Annamalai University, Annamalaiagar, Tamil Nadu-608 002, India.

E-mail:

paribala@sancharnet.in

decreasing levels of high-density lipoprotein (HDL), an antiatherogenic lipid [4]. Insulin therapy affords effective glycaemic control, yet its short comings such as ineffectiveness on oral administration, short shelf life, requirement of constant refrigeration and, in the event of excess dosage, fatal hypoglycaemia, limit its usage. Treatment with sulphonylureas and biguanides is also associated with side-effects [5].

For various reasons, in recent years the popularity of herbal medicine has increased. Dietary measures and traditional plant therapies as prescribed by Ayurvedic and other indigenous systems of medicine were commonly used in India [6]. Plant drugs are frequently considered to be less toxic and more free from side-effects than synthetic ones [7]. World Health Organization (WHO) has also recommended the evaluation of the effectiveness of plant in conditions where we lack safe modern drugs [8,9].

In the traditional system of Indian Medicine Plant Formulation and several cases, combined extracts of plants are used as the drug of choice rather than individual. Many of these have shown promising effects [10]. Various herbal formulations like D-400 [11], Trasina [12], Diamed [13], Cogent db [14] are well known for their antidiabetic actions.

Cogent db, a novel and unique polyherbal drug produced by Cybele Herbal Laboratories, Kerala, India, for diabetes mellitus, contains the extracts of nine medicinal plants whose composition and concentrations are given in table 1. There is no report available in the literature for the antihyperlipidaemic effect of Cogent db in alloxan-induced diabetic rats. The present investigation was undertaken to study the effect of Cogent db, a crude drug on changes in blood glucose, plasma insulin, activity of lipogenic enzymes, serum and tissue (liver and kidney) lipids. The effect of this drug on different parameters were compared with glibenclamide.

Table 1 Composition of Cogent db tablet

Botanical name	Family	Weight (mg)
<i>Azadirachta indica</i>	Meliaceae	360
<i>Phyllanthus emblica</i>	Euphorbiaceae	85
<i>Terminalia bellerica</i>	Combretaceae	85
<i>Terminalia chebula</i>	Combretaceae	85
<i>Tribulus terrestris</i>	Zygophyllaceae	120
<i>Trigonella foenum graecum</i>	Apiaceae	15
<i>Curcuma longa</i>	Zingiberaceae	95
<i>Syzygium cumini</i>	Myrtaceae	240
<i>Rotula aquatica</i>	Lythiraceae	120

Research Design and Methods

Animals

Male Wistar rats body weight of 180–200 g were obtained from Central Animals House, Raja Muthiah Medical College, Annamalai University. The animals were fed on standard pellet diet (Hindustan Lever, India) and water *ad libitum*.

Test Drugs and Chemicals

Cogent db, a herbal drug was a gift from Cybele Herbal Laboratories, Kerala, India. Alloxan monohydrate was purchased from BDH chemicals Ltd, Poole, England. Radioimmuno assay kit for insulin assay was purchased from Boehringer Mannheim, Germany. All other biochemicals and chemicals used were of analytical grade.

Drug Administration

Cogent db tablets were finely powdered, suspended in distilled water and administered orally through orogastric tube at a dose 0.45 g/kg body weight [14].

Experimental Induction of Diabetes in Rats

The rats were injected intraperitoneally with alloxan monohydrate dissolved in sterile normal saline at a dose of 150 mg/kg body weight [15]. After 2 weeks, rats with moderate diabetes having glycosuria (indicated by Benedict's qualitative test) and hyperglycaemia (i.e. with a blood glucose of 200–260 mg/dL) was used for the experiment.

Experimental Design

In the experiment, a total of 24 rats (18 diabetic surviving rats, six normal rats) were used. Diabetes was induced in rats 2 weeks before starting the experiment. The rats were divided into four groups after the induction of alloxan diabetes. In the experiment, six rats were used in each group.

Group 1 – Normal rats given 2 ml of physiological saline;

Group 2 – Diabetic control rats given 2 ml of physiological saline;

Group 3 – Diabetic rats given aqueous solution of Cogent db (0.45 g/kg body weight) daily using an intragastric tube for 40 days;

Group 4 – Diabetic rats given aqueous solution of glibenclamide (600 µg/kg body weight) daily using an intragastric tube for 40 days.

At the end of 40 days, the animals were deprived of food overnight and sacrificed by decapitation. Blood was collected in tubes containing potassium oxalate and sodium fluoride solution for the estimation of blood glucose. Plasma was separated for the estimation of insulin. Liver and kidney was dissected out, washed in ice-cold saline, patted dry and weighted.

Analytical Methods

Fasting blood glucose was estimated by *O*-toluidine method [16]. The plasma insulin levels in normal and experimental rats were estimated by using radioimmunoassay kit (Boehringer Mannheim, Germany). Glucose-6-phosphate dehydrogenase, and malic enzyme was determined using the method of Baquer *et al.* [17]. Protein concentration was determined by the method of Lowry *et al.* [18]. Lipids were extracted from tissues by the method of Folch *et al.* [19]. Serum and tissue total cholesterol [20], triglycerides [21], phospholipids [22], free fatty acids in tissues [23], serum HDL-cholesterol [24] and serum low-density lipoproteins (LDL) [25] were estimated by standard procedures. Serum very-low-density lipoprotein was calculated according to Friedewald *et al.* [25].

Statistical Analysis

Statistical analysis of the results performed with student's *t*-test. Results are presented as mean ± S.D.

Results

Table 2 shows the levels of fasting blood glucose, plasma insulin and the activities of the two hepatic lipogenic

enzymes glucose-6-phosphate dehydrogenase and malic enzyme of normal and experimental animals. There was a significant increase in blood glucose and decrease in plasma insulin and hepatic lipogenic enzymes activity in diabetic rats compared with normal rats. Administration of Cogent db and glibenclamide tends to bring the values to near normal.

Table 3 shows the levels of serum total cholesterol, triglycerides, phospholipids high-density lipoprotein (HDL), low-density lipoprotein (LDL) and very-low-density lipoprotein (VLDL) of normal and experimental animals. In diabetic rats, serum lipids were significantly increased where as HDL-cholesterol was significantly decreased compared with normal rats. Treatment with Cogent db and glibenclamide significantly reversed these effects compared with diabetic rats.

Table 4 shows the content of total cholesterol, triglycerides, phospholipids and free fatty acids in liver and kidney of normal and experimental animals. Total cholesterol, triglycerides, phospholipids and free fatty acids content in tissues were significantly increased in diabetic animals compared with normal. Whereas in rats treated with Cogent db and glibenclamide showed significant decrease in tissue lipids compared with corresponding diabetic control rats.

Discussion

The diabetogenic effects of alloxan attributed to a specific cytotoxic action mediated by hydroxy radical generation on pancreatic β-cells. This damages a large number of β-cells, resulting in a decrease in endogenous insulin release. Alloxan administered rats therefore become hyperglycaemic in a short period of time, followed by a hepatic glucose overproduction [26].

In our study, we have observed a significant increase in plasma insulin when alloxan diabetic rats treated

Table 2 Blood glucose, plasma insulin, activities of glucose-6-phosphate dehydrogenase and malic enzyme of normal and experimental groups

Groups	Fasting blood glucose (mg/100 ml)	Plasma insulin (µU/ml)	Glucose-6- phosphate dehydrogenase (U/g wet tissue)	Malic enzyme (U/g wet tissue)
Normal	75.45 ± 5.3	25.3 ± 2.5	4.80 ± 0.21	1.77 ± 0.07
Diabetic control	230.5 ± 13.6***	11.1 ± 1.0***	2.26 ± 0.14***	0.81 ± 0.10***
Diabetic + Cogent db (0.45 g/kg)	83.4 ± 6.7 ^{ccc}	22.6 ± 2.1 ^{ccc}	4.20 ± 0.24 ^{ccc}	1.51 ± 0.05 ^{ccc}
Diabetic + glibenclamide (600 µg/kg)	91.2 ± 6.9 ^{ccc}	20.8 ± 1.9 ^{ccc}	4.30 ± 0.20 ^{ccc}	1.53 ± 0.06 ^{ccc}

Values are given as mean ± SD for six rats in each group.

Diabetic control was compared with normal, ****p* < 0.001; experimental groups were compared with diabetic control, ^{ccc}*p* < 0.001.

U = 1 µmol of NADPH formed per g of fresh weight of liver per minute at 37 °C.

Table 3 Serum lipid profile of normal and experimental groups

Groups	Total cholesterol	Triglycerides	Phospholipids	HDL-cholesterol	LDL-cholesterol	VLDL-cholesterol
Normal	89 ± 4.6	83.9 ± 6.1	112.1 ± 6.3	51.2 ± 4.0	21.0 ± 1.8	16.8 ± 1.3
Diabetic control	210 ± 7.9***	160.5 ± 8.2***	225.2 ± 8.4***	28.5 ± 2.2***	149.4 ± 7.3***	32.1 ± 1.6***
Diabetic + Cogent db (0.45 g/kg)	110 ± 6.2†††	115.5 ± 5.8†††	179.1 ± 6.5†††	49.2 ± 3.7†††	37.9 ± 3.6†††	23.1 ± 1.2†††
Diabetic + glibenclamide (600 µg/kg)	121 ± 6.7†††	120.5 ± 6.5†††	188.2 ± 7.5†††	46.2 ± 3.1†††	52.1 ± 4.7†††	24.1 ± 1.3†††

Values are expressed as mg/dL serum.

Values are given as mean ± SD for six rats in each group.

Diabetic control was compared with normal, ****p* < 0.001; experimental groups were compared with diabetic control, †††*p* < 0.001.

with Cogent db. The possible mechanism by which Cogent db brings about its antidiabetic effect could be due to potentiation of the insulin effect of plasma by increasing the pancreatic secretion of insulin from existing β-cells of islets of Langerhans or decreased extraction of insulin by the liver [14].

Liver is regarded as one of the central metabolic organ in the body, regulating and maintaining homeostasis. It is an insulin-dependent tissue, which plays a pivotal role in glucose and lipid homeostasis and is severely affected during diabetes. Diabetes results in a decrease in glucose utilization and an increase in glucose production in insulin-dependent tissues such as liver [27]. The liver tissue participates in the uptake, oxidation of free fatty acids, synthesis of cholesterol and phospholipids and secretion of specific classes of plasma lipoproteins [28].

During diabetes, the hepatic lipogenesis is decreased and lipolysis is increased [29]. Reversal of the two hep-

atic lipogenic enzymes glucose-6-phosphate dehydrogenase and malic enzyme in diabetic liver to near normal by Cogent db treatment suggests that the hydrogen shuttle systems and the redox state of the cell becomes more oxidized. The above increases the formation of NADPH for increased utilization in lipogenesis and, in turn, activating the lipogenic enzymes, as NADPH is a strong inhibitor of glucose-6-phosphate dehydrogenase [30]. The altered total lipid profile in the liver during diabetes is substantially restored by Cogent db treatment, suggesting that the improved lipogenic transhydrogenase reactions by Cogent db treatment may provide increased amounts of NADPH for the reversal of the biosynthesis of fatty acids to almost control levels. The NADPH-producing enzyme glucose-6-phosphate dehydrogenase also showed a decrease in activity in experimental diabetes in red blood cells (RBC), which was restored to control levels by insulin treatment [31].

Table 4 Tissue lipids (liver and kidney) of normal and experimental groups

Groups	Total cholesterol	Triglycerides	Phospholipids	Free fatty acids
Normal				
Liver	5.67 ± 0.24	3.51 ± 0.28	16.38 ± 1.56	7.14 ± 0.65
Kidney	8.13 ± 0.52	4.25 ± 0.35	14.32 ± 1.25	7.82 ± 0.62
Diabetic control				
Liver	8.45 ± 0.35***	6.72 ± 0.39***	32.53 ± 2.93***	12.35 ± 0.97***
Kidney	11.42 ± 0.65***	7.36 ± 0.53***	30.43 ± 2.31***	15.34 ± 1.42***
Diabetic + Cogent db (0.45 g/kg)				
Liver	6.13 ± 0.28†††	3.95 ± 0.27†††	16.12 ± 1.58†††	8.16 ± 0.57†††
Kidney	7.36 ± 0.56†††	4.83 ± 0.31†††	15.28 ± 1.41†††	3.21 ± 0.73†††
Diabetic + glibenclamide (600 µg/kg)				
Liver	6.65 ± 0.31†††	4.43 ± 0.31†††	16.94 ± 1.63†††	8.52 ± 0.48†††
Kidney	7.84 ± 0.64†††	5.12 ± 0.34†††	16.37 ± 1.57†††	9.81 ± 0.87†††

Values are expressed as mg/g wet tissue.

Values are given as mean ± SD for six rats in each group.

Diabetic control was compared with normal, ****p* < 0.001; experimental groups were compared with diabetic control, †††*p* < 0.001.

In non-diabetic subjects, insulin suppresses the release of VLDL 1 apo B from liver. In type 2 diabetes, however, the production of this VLDL 1 apo B is not impaired, which may be one of the root causes of characteristic hypertriglyceridaemia [32]. The release of large triglyceride-rich VLDL subfractions from the liver leads to the production of small LDL, a particle strongly associated with the development of atherosclerosis due in part to its increased susceptibility to oxidation [33].

The altered composition of lipoproteins in diabetes is facilitated by the action of cholesterol ester transfer protein (CETP) [34]. Triglyceride-rich particles have a prolonged residence time within the circulation and are exposed to the action of this enzyme for a longer period of time, the consequence of which is the net transfer of triglyceride from VLDL to HDL and LDL, with cholesterol moving in the opposite direction [35].

In our study, we have also observed higher levels of cholesterol, triglycerides, phospholipids and free fatty acids in serum and tissues (liver and kidney) of alloxan diabetic rats. Similar results have been reported by other workers in alloxan diabetic rats [36,37].

The abnormalities in lipid metabolism generally lead to elevation in the levels of serum lipids and lipoproteins, which in turn play an important role in the occurrence of premature and severe atherosclerosis, which affects patients with diabetes mellitus. The development of diabetic nephropathy contributes an additional cause of plasma lipid abnormalities [38], and microalbuminuria (MAU) is associated with the development of clinical proteinuria, chronic renal failure and premature cardiovascular mortality in diabetics [39].

Lowering of serum lipids concentration through dietary or drugs thereby seems to be associated with a decrease in the risk of vascular disease [40]. The abnormal high concentration of serum lipids in diabetic subjects is mainly due to increased mobilization of free fatty acids from the peripheral fat depots, as insulin inhibits the hormone-sensitive lipase. However, glucagons, catecholamines and other hormones enhance lipolysis. The marked hyperlipaemia that characterized the diabetic state may therefore be regarded as a consequence of the uninhibited actions of lipolytic hormones on the fat deposits [41]. The levels of lipids in serum and tissues were brought near normal by the treatment of Cogent db in alloxan diabetic rats. The effect of Cogent db on diabetic hyperlipaemia could be through its control of hyperglycaemia. This is agreement with the facts that: (1) the level of glycaemic control is the major determinant of total and very-low-density lipoprotein (VLDL) triglyceride concentrations [42]; and (2) improved glycaemic control following sulphonylurea therapy decreases the

levels of serum VLDL and total triglycerides [43].

In the present study, Cogent db has been shown to have a significant hypolipidaemic effect in alloxan diabetic rats. This could be correlated with a previous study that reported that *Trigonella foenum graecum* [44], *Curcuma longa* [45], *Azadirachta indica* [46], *Syzygium cumini* [47] (constituents of Cogent db) have hypolipidaemic effect in diabetic animals.

The synthetic oral hypoglycaemic agents can produce a series of side-effects including haematological, gastrointestinal reactions, hypoglycaemic coma and disturbances of liver and kidney [48]. Through this study, we can conclusively state that Cogent db, a polyherbal drug, has strong antihyperlipidaemic effect. This could be due to different types of active principles, each with a single or a diverse range of biological activities. It can serve as a good adjuvant in the present armamentarium of antidiabetic drugs.

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